作者公开网页的浏览器导出（不是作者原始PDF） • https://shufengc.dev/ • 2026-10-24

M.S. EE AT COLUMBIA · SYSTEMS & SECURITY RESEARCH · PREV. STRIPE

Shufeng (Alex) Chen

I work on the reliability and correctness of GPU software at Columbia's ICE Lab: coverage-guided fuzzing, divergence-aware program analysis, and sanitizer-guided triage for CUDA libraries. Most recently I built backend infrastructure at Stripe, shipping self-serve onboarding APIs for its SFTP file-transfer platform.

View Research View Experience



RECENT UPDATES

News

Aug 2026	Completed a software engineering internship at Stripe on the SFTP file-transfer platform in Cash & Banking Infrastructure.
Aug 2026	Finishing a paper on CUDA library fuzzing with thread-granular, divergence-sensitive coverage feedback.
Spring 2026	Admitted to the MS EE/CE Honors Program at Columbia.
Feb 2026	Joined the ICE Lab at Columbia as a Graduate Research Assistant working on GPU fuzzing.
Aug 2025	Started the M.S. in Electrical Engineering at Columbia University as a Tesla Scholar, awarded to the top 5% of admitted students.

SYSTEMS & SECURITY

Research

Graduate Research Assistant

Feb 2026 – Present

ICE Lab · Columbia University

My current focus is coverage-guided fuzzing for GPU software. Traditional edge coverage treats a CUDA kernel launch as one event and misses the SIMT execution states where GPU bugs live, so we built a fuzzer that tracks coverage at the granularity of individual threads, rewards divergence-sensitive behavior, and replays

corpora under compute-sanitizer to detect and triage memory-safety defects in real CUDA libraries and benchmark suites.

- Built coverage-guided fuzzing infrastructure for CUDA targets: NVBit-based coverage engines integrated with AFL++, persistent-mode harnesses, and fault isolation for poisoned CUDA contexts.
- Root-caused five memory-safety defects, four original and one a bypass of a deployed upstream fix.
- Repaired and classified a 39-target CUDA benchmark fuzzing suite used for evaluation.
- Ran a three-arm coverage evaluation on A100 GPUs showing the divergence-aware representation sustains new-coverage discovery after plain edge coverage saturates.

Ongoing work extends the pipeline toward agentic program analysis and repair for GPU code.

CUDA NVBit AFL++ compute-sanitizer cuda-gdb C/C++ Python

RESEARCH OUTPUT

Publications

IN PREPARATION

THREADSIGHT: CUDA Library Fuzzing with Thread-Granular, Divergence-Sensitive and Sanitizer-Guided Coverage Feedback

In preparation · 2026 · title tentative

PUBLISHED

Global-local Dynamic Adversarial Learning for Cross-domain Sentiment Analysis

Mathematics 11(14):3130 · 2023

Juntao Lyu, Zheyuan Zhang, **Shufeng Chen**, Xiying Fan*.

MDPI

Breast Cancer Detection Model Training Strategy Based on Continual Learning

CAIBDA 2022 · Conference Paper

Shufeng Chen, Fenghui Tang.

IEEE Xplore

CAREER JOURNEY

Experience

Software Engineer Intern

May 2026 — Aug 2026

Stripe · Cash & Banking Infrastructure · Seattle, WA

- Designed and shipped self-serve onboarding APIs for Stripe's SFTP file-transfer platform: a new gRPC configuration service whose three operations replace a multi-PR, deploy-and-wait workflow with a single validated API call, reducing partner onboarding from days to hours.
- Built the configuration validation layer behind the onboarding APIs and rolled it out behind feature flags with no behavior change to live integrations.

- Shipped a Java gRPC service that captures partner SSH host keys during the connection handshake, removing per-partner infrastructure changes and manual log searches.

Java gRPC Protocol Buffers SFTP / SSH Feature Flags

Software Engineer Intern

Jul 2025 – Aug 2025

Tencent · Cloud Computing Group

- Built backend microservices for distributed inference pipelines on Kubernetes-managed HPC clusters; automated container lifecycle & remote job operations for smoother orchestration.
- Developed internal cluster management and scheduling APIs (Spring Boot/Security, Redis/MySQL) and integrated cloud metadata synchronization for lifecycle tracking.
- Improved ingestion/telemetry services by shipping a CLI-based pipeline adopted by 20+ internal apps; enabled high-throughput streaming from 4,300+ edge nodes.
- Deployed pre-embedding pipelines across Tencent Cloud and AWS (EKS/EC2), improving inference latency by 1.3× and reducing CPU load by 2.2× via caching and preprocessing.

Kubernetes Spring Boot Redis MySQL AWS EKS Docker

Full Stack Software Engineer Intern

Sep 2024 – Nov 2024

Conflux Foundation · Tree-Graph Research Institute

- Built a cross-platform crypto analytics app (React Native) with GPT-based insights; summarized 5,000+ news sources using LDA topic modeling.
- Designed an automated ETL pipeline (Scrapy, AWS Lambda, DynamoDB, EventBridge) and indexed real-time data to OpenSearch; improved search API performance by 33%.
- Shipped high-performance secrets-detection microservices (Go/Python/Rust) over gRPC + Kubernetes; reduced P95 latency from 170ms to 40ms and supported 10K+ QPS using Bloom-filter warm cache.
- Implemented serverless remediation workflows (Step Functions + Lambda) and production observability (Prometheus/Grafana, structured logs).

FEATURED WORK

Projects

Selected repos from github.com/shufengc

Distributed Key-Value Store

GitHub

Go gRPC Multi-Raft MVCC

Horizontally scalable, fault-tolerant key-value storage system featuring strong consistency and distributed transactions.

- Raft Consensus & Sharding: Implemented Multi-Raft for log replication, dynamic region splitting, and leader transfer without downtime.
- Global Scheduler: Developed a heartbeat-driven scheduler to monitor cluster metadata and auto-rebalance replicas across nodes.
- Percolator 2PC: Built distributed transaction support with Snapshot Isolation handling prewrites, commits, and rollbacks.

QuantHarbor

GitHub

Python FastAPI React RAG

End-to-end AI financial research platform transforming unstructured market documents into citation-grounded insights.

- Multi-Agent Architecture: Orchestrated specialized agents (Data Collector, Analyzer, Report Generator, Deep Search) collaborating in a shared variable space.
- RAG Pipeline: Built a document-to-insight pipeline integrating PDF ingestion, vector indexing, and citation tracing for verifiable insights.
- VLM Feedback Loops: Implemented built-in vision agents that automatically correct chart issues during publication-grade report generation.

- Concurrency Control: Managed concurrent operations via MVCC and per-key latching on an embedded LSM-tree engine.

- Interactive UI: Delivered a full-stack dashboard for system configuration, execution monitoring via WebSockets, and checkpoint/resume management.

Patton Food

GitHub

Java Spring Boot React PostgreSQL

Full-stack, cloud-deployed online food ordering web application with clean API design and a modern UI.

- User registration and login with session-based authentication using Spring Security.
- Password encryption with BCrypt and secure session management.
- Restaurant/menu browsing with cart and checkout flow.
- Cloud deployment on AWS with production-ready configuration.

PDF-AI: Conversational Q&A System

GitHub

TypeScript React LangChain OpenAI

AI assistant that lets users upload PDFs and ask natural language questions about their content.

- PDF upload & processing pipeline with chunking and vector indexing.
- Retrieval + LLM answering with conversation context and source citations.
- Responsive chat UI with streaming responses.
- Latency optimizations via caching and request shaping.

L2 Game Engine

GitHub

C++ Lua SDL Box2D

Cross-platform 2D game engine in C++ with Lua scripting — designed to keep the runtime fast, modular, and easy to extend.

LinguAR: AR Language Learning

Demo

Unity C# ARKit GPT-4o

iOS AR app prototype combining real-scene recognition, translation, and multimodal tutoring for daily language practice.

- C++ core runtime: real-time game loop, scene/actor system, and engine APIs optimized for iteration speed.
- Lua scripting + language hosting: externalized gameplay logic with script-facing APIs for actor creation and behaviors.
- Box2D physics: collision handling, rigid body dynamics, friction, and gravity for realistic 2D gameplay.
- Shipped a complete game on the engine: Havoc: 300-Seconds Escape.

- Built the end-to-end AR learning loop: object recognition (YOLOX) → translation overlay (Google Cloud Translate) → pronunciation output (AWS Polly).
- Led AR shooting mini-game feature implementation: fixed screen flicker, added SFX/VFX + scoreboard.
- Integrated voice-first AI tutoring (Whisper STT + GPT-4o + TTS) for conversational practice with multimodal I/O.
- Delivered a usability-tested team prototype with story maps and iterative refinements.

INTERACTIVE WORK

Game & XR Projects

VR Room Simulator

Unreal Engine · Blueprints

VR walkthrough simulator focused on interaction and environment fidelity, built for real-time performance.

- Interactive VR navigation with user-friendly scene flow
- Blueprint-driven interaction logic for rapid iteration
- Performance-oriented real-time rendering considerations

Demo Video

A2 GO! (AR Location-Based Game)

Unity · C# · Mapbox

AR experience inspired by location-based gameplay, combining map navigation with AR interactions.

- Map-based discovery using Mapbox integration
- AR interactions and UI flows for collection/progression
- Mobile-first performance and interaction design

Demo Video

RENOVAR (AR Furniture Preview)

Unity · XR/AR · Meta Quest

AR prototype for previewing furniture in space before purchase, emphasizing scale, placement, and usability.

- Virtual placement workflow for in-room visualization
- Interaction and UI patterns optimized for XR devices
- End-to-end prototype delivery with structured iteration

Project Page

The Legend of Zelda (Unity Remaster)

Unity · C#

Unity remaster-style gameplay prototype focusing on core action-adventure mechanics and moment-to-moment feel.

- Implemented key gameplay systems (combat/items/progression)
- Enemy behavior and encounter tuning for gameplay clarity
- Polished controls and UI feedback loops

Play

Echo Space (Echolocation Adventure)

Unity · C#

Gameplay prototype built around echolocation-style exploration, turning sound-driven feedback into navigation.

- Core mechanic: reveal environment state through echolocation cues
- Designed exploration loop to support discovery and tension
- Implemented interaction and pacing to keep sessions engaging

Play

Chrono Portal: A Journey Home

Unity · C#

Puzzle game centered on portal mechanics, emphasizing level logic, item interactions, and UI clarity.

- Portal-based puzzle design with staged difficulty progression
- Inventory UI and interaction system for puzzle tooling
- Shipped playable build with multiple levels and clear objectives

Play

Let's build something great together

I'm always happy to talk about systems, security, and interesting engineering problems.

Academic: sc5739@columbia.edu
Industry: shufeng.c.dev@gmail.com

© 2026 Shufeng Chen. All rights reserved.